The Systemic Market Fragility Index (SMFI): Detecting Limit Order Book Manipulation via Signature-Based Zigzag Persistent Homology

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Abstract

Topological methods have already been shown to capture exploitable structure in financial data—at the level of correlation networks (Turiel, Barucca & Aste, 2022), macro index dynamics (Gidea & Katz, 2018), and, most closely, within the limit order book itself for price forecasting (HLOB; Briola, Bartolucci & Aste, 2024). What remains open is whether a topological construction operating natively at LOB-event granularity can **detect manipulation**—not just structure or forecast price—with the statistical rigor (FDR, lead-time) that a surveillance tool requires, and whether that construction can be extended beyond a single instrument to genuinely systemic, cross-asset manipulation patterns. This paper proposes the **Systemic Market Fragility Index (SMFI)**: zigzag persistent homology applied to log-signature embeddings of LOB data, benchmarked against volume-based (VPIN, OFI) and structural (motif/graph-based) detectors on False Discovery Rate and detection lead-time; extended to a genuinely systemic, multi-asset setting via a bounded multiparameter persistence construction that tracks topological coupling *between* correlated instruments, not just within one; and stress-tested via a **generalized constrained adversarial evaluation protocol** (CAEP)—a reusable framework for testing any market-surveillance detector against an adaptive, economically-anchored adversary—instantiated here against a reservoir-computing generator.

1 Core Falsifiable Hypotheses

Primary Hypothesis (Single-Asset Detection): *Zigzag persistent homology applied to the log-signatures of Limit Order Book (LOB) data detects structural liquidity collapse with a lower False Discovery Rate (FDR) and earlier detection lead-time than Volume-Synchronized Probability of Informed Trading (VPIN), Order Flow Imbalance (OFI), quote-to-trade ratio, and at least one structural (non-volume-based) 2024–2025 baseline, during large-scale parameterized heuristically injected spoofing events, and generalizes qualitatively to real, prosecuted historical manipulation cases.*

Secondary Hypothesis (Systemic, Cross-Asset Extension): *A bounded multiparameter zigzag persistence construction, tracking topological coupling between a small cluster of correlated instruments over time, detects coordinated cross-asset manipulation episodes (price-linked spoofing across correlated tickers) that single-asset SMFI, run independently on each ticker, misses or detects later.*

2 Related Work

2.1 Topological Methods in Financial Microstructure

Persistent homology has established systemic risk signals on correlation networks (Turiel et al., 2022) and crash indicators (Gidea & Katz, 2018). HLOB (Briola et al., 2024) applied homological convolutions for price prediction. SMFI differentiates itself by targeting continuous-event manipulation detection through rough-path log-signatures and zigzag persistence.

2.2 Structural Spoofing Baselines

SMFI is benchmarked against current structural architectures: Zheng (2025) temporal motif GNNs, Lin & Yang (2025) contrastive representations, and Fabre & Challet (2025) Hawkes spoofability models, moving beyond traditional volume metrics (VPIN, OFI).

2.3 Adversarial Evaluation Precedents

While Cartea et al. (2023) demonstrated adaptive spoofing agents, SMFI introduces the Generalized Constrained Adversarial Evaluation Protocol (CAEP) to formally evaluate surveillance robustness subject to an analytical economic-content floor $M(\cdot) \geq \tau$ derived from Cartea, Jaimungal & Wang (2020) and Tao et al. (2020).

3 Three-Tier Validation Framework

3.1 Tier 1: Parameterized Heuristic Injections

Continuous Poisson-process cancels and multi-level depth layering (1–8 levels) parameterized by analytical distortion models. To ensure statistical power without data purchase, FI-2010 is merged into the primary LOBSTER corpus to ensure $N_{\text{inj}} \geq 200$ independent 35-second windows (Audit Blocker D1). A dedicated subset of $\sim$30 injections applies simultaneous perturbations across a 3-ticker cluster to test cross-asset coupling (Audit A2).

3.2 Tier 2: Historical Out-of-Sample Validation

Replay across the 2010 Flash Crash, 2015 Coscia case, and Do & Putniņš (2023) prosecuted enforcement events with zero data leakage from Tier 1.

3.3 Tier 3: CAEP Adversarial Stress Test

The adversarial min-max loop minimizes surveillance detection score subject to economic manipulation constraints:

$$\min_{\theta} \mathbb{E}_z \left[\text{SMFI}(G_\theta(z))\right] \quad \text{subject to} \quad M(G_\theta(z)) \geq \tau \tag{1}$$

4 Mathematical Methodology (Single-Asset Core)

4.1 Log-Signatures over Microsecond Streams

Paths of mid-price, depth-weighted imbalance, and top-N volumes are embedded into truncated log-signatures at depth $N = 4$.

4.2 Vietoris-Rips Complexes and Zigzag Persistence

Simplicial complexes are built to the 2-skeleton on sliding windows of W vectors with radius ε. As mandated by Audit M3, $W \in \{5, 10, 20, 50\}$ is calibrated jointly with ε by maximizing β_0 dynamic range on clean windows. Consecutive complexes $C(t_k)$ and $C(t_{k+1})$ are linked by inclusions into their union:

$$C(t_k) \hookrightarrow C(t_k) \cup C(t_{k+1}) \hookleftarrow C(t_{k+1}) \tag{2}$$

permitting non-monotonic appearance/disappearance tracking. Primary signal resides in β_0 (transient disconnected order clusters).

4.3 Lifespan Filtering and Fragility Signal

Bars with lifespan below the 10th percentile empirical noise floor δ are pruned. The SMFI signal is computed as the 2-Wasserstein distance between consecutive persistence diagrams:

$$\mathrm{SMFI}(t) = W_2(D_t, D_{t-1}) \tag{3}$$

4.4 Benjamini-Hochberg FDR Protocol (Audit M4)

Alarms trigger at threshold θ (95th percentile clean null). The per-event p-value is rigorously defined via a one-sided empirical permutation test:

$$p_i = \frac{\#\{\text{clean windows } j : \max_{t\in[-5\mathrm{s},+30\mathrm{s}]} \mathrm{SMFI}_j(t) > \max_{t\in[-5\mathrm{s},+30\mathrm{s}]} \mathrm{SMFI}_i(t)\}}{N_{\text{clean}}} \tag{4}$$

Ranked p-values are corrected at target FDR $q = 0.05$.

5 Multiparameter Cross-Asset Extension

For $k = 3$ correlated tickers, asset-coupling simplicial complexes are formed where edges denote inter-asset signature distance $< \varepsilon$. Evolving through time, this forms a 2-parameter filtration summarized via the discrete rank invariant to bypass EXPSPACE complexity.

6 Datasets and Pipeline Governance

- **LOBSTER Sample + FI-2010 (Merged):** Tier 1 corpus guaranteeing $N_{\text{inj}} \geq 200$.
- **Binance API:** Stress-testing isolated via `SnapshotToEventAdapter` (Audit D2).
- **Topological Discrepancy Halt Criterion:** Dionysus2 and GUDHI must agree within $\varepsilon_{\text{pd}} = 0.01$; $> 5\%$ violations halt execution immediately.
- **Total Compute Budget:** 140 GPU-hours (Total program footprint: 250 GPU-hours).

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